

UNIT-5 | Lecture 1

Evaluation of Relational Algebra Expressions

Coverage: Hour 1

Learning Objectives

Understand how relational algebra expressions are evaluated.

Learn operator trees and evaluation order.

Relational Algebra Expression

A relational algebra expression is a sequence of operations that produces a relation as its result.

Example: π Name (σ Branch='CSE' (Student)).

Evaluation Process

Start with the base relations.

Apply selection and other unary operations to reduce data where possible.

Apply binary operations such as join, union, difference, or product.

Apply projection and produce the final result.

Operator Tree

An **operator tree** represents the relational algebra expression graphically. Leaf nodes are base relations and internal nodes are relational operators.

For π Name(σ Branch='CSE'(Student)), the tree is conceptually: **Projection** $\rightarrow$ **Selection** $\rightarrow$ **Student**.

Materialization vs Pipelining

Materialization: Store the intermediate result before executing the next operation.

Pipelining: Pass tuples directly from one operation to the next without storing the complete intermediate relation.

Important Point

Evaluation should try to reduce the size of intermediate results because smaller intermediate relations usually require less memory and processing.

Key Takeaway

Evaluating a relational algebra expression means executing its operators in an effective order to produce the required result.